

Milestone 3 – Papers Submission for Approval

BDM 3035 – Big Data Capstone Project

Group 1

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1. Nathani, D., Madaan, L., Roberts, N., Bashlykov, N., Menon, A., Moens, V., Budhiraja, A., Magka, D., Vorotilov, V., Chaurasia, G., Hupkes, D., Cabral, R. S., Shavrina, T., Foerster, J., Bachrach, Y., Wang, W. Y., & Raileanu, R. (2025, February 20). *MLGYM: A new framework and benchmark for Advancing AI research Agents*.

arXiv.org. <https://arxiv.org/abs/2502.14499>

Abstract: We introduce Meta MLGym and MLGym-Bench, a new framework and benchmark for evaluating and developing LLM agents on AI research tasks. This is the first Gym environment for machine learning (ML) tasks, enabling research on reinforcement learning (RL) algorithms for training such agents. MLGym-bench consists of 13 diverse and open-ended AI research tasks from diverse domains such as computer vision, natural language processing, reinforcement learning, and game theory. Solving these tasks requires real-world AI research skills such as generating new ideas and hypotheses, creating and processing data, implementing ML methods, training models, running experiments, analyzing the results, and iterating through this process to improve on a given task. We evaluate a number of frontier large language models (LLMs) on our benchmarks such as Claude-3.5-Sonnet, Llama-3.1 405B, GPT-4o, o1-preview, and Gemini-1.5 Pro. Our MLGym framework makes it easy to add new tasks, integrate and evaluate models or agents, generate synthetic data at scale, as well as develop new learning algorithms for training agents on AI research tasks. We find that current frontier models can improve on the given baselines, usually by finding better hyperparameters, but do not generate novel hypotheses, algorithms, architectures, or

substantial improvements. We open-source our framework and benchmark to facilitate future research in advancing the AI research capabilities of LLM agents.

2. Hong, S., Zhuge, M., Chen, J., Zheng, X., Cheng, Y., Zhang, C., Wang, J., Wang, Z., Yau, S. K. S., Lin, Z., Zhou, L., Ran, C., Xiao, L., Wu, C., & Schmidhuber, J. (2023, August 1). *MetaGPT: Meta Programming for a Multi-Agent Collaborative Framework*. arXiv.org. <https://arxiv.org/abs/2308.00352>

Abstract: Remarkable progress has been made on automated problem solving through societies of agents based on large language models (LLMs). Existing LLM-based multi-agent systems can already solve simple dialogue tasks. Solutions to more complex tasks, however, are complicated through logic inconsistencies due to cascading hallucinations caused by naively chaining LLMs. Here we introduce MetaGPT, an innovative meta-programming framework incorporating efficient human workflows into LLM-based multi-agent collaborations. MetaGPT encodes Standardized Operating Procedures (SOPs) into prompt sequences for more streamlined workflows, thus allowing agents with human-like domain expertise to verify intermediate results and reduce errors. MetaGPT utilizes an assembly line paradigm to assign diverse roles to various agents, efficiently breaking down complex tasks into subtasks involving many agents working together. On collaborative software engineering benchmarks, MetaGPT generates more coherent solutions than previous chat-based multi-agent systems.

3. Liu, Z., Zhang, Y., Li, P., Liu, Y., & Yang, D. (2023, October 3). *A dynamic LLM-Powered agent network for Task-Oriented agent collaboration*. arXiv.org. <https://arxiv.org/abs/2310.02170>

Abstract: Recent studies show that collaborating multiple large language model (LLM) powered agents is a promising way for task solving. However, current approaches are constrained by using a fixed number of agents and static communication structures. In

this work, we propose automatically selecting a team of agents from candidates to collaborate in a dynamic communication structure toward different tasks and domains. Specifically, we build a framework named Dynamic LLM-Powered Agent Network (**DyLAN**) for LLM-powered agent collaboration, operating a two-stage paradigm: (1) Team Optimization and (2) Task Solving. During the first stage, we utilize an agent selection algorithm, based on an unsupervised metric called Agent Importance Score, enabling the selection of best agents according to their contributions in a preliminary trial, oriented to the given task. Then, in the second stage, the selected agents collaborate dynamically according to the query. Empirically, we demonstrate that DyLAN outperforms strong baselines in code generation, decision-making, general reasoning, and arithmetic reasoning tasks with moderate computational cost. On specific subjects in MMLU, selecting a team of agents in the team optimization stage improves accuracy by up to 25.0% in DyLAN.

4. Wang, J., Jiang, R., Yang, C., Wu, Z., Onizuka, M., Shibasaki, R., Koshizuka, N., & Xiao, C. (2024, February 22). *Large Language Models as Urban Residents: An LLM Agent Framework for Personal Mobility Generation*.

arXiv.org. <https://arxiv.org/abs/2402.14744>

Abstract: This paper introduces a novel approach using Large Language Models (LLMs) integrated into an agent framework for flexible and effective personal mobility generation. LLMs overcome the limitations of previous models by effectively processing semantic data and offering versatility in modeling various tasks. Our approach addresses three research questions: aligning LLMs with real-world urban mobility data, developing reliable activity generation strategies, and exploring LLM applications in urban mobility. The key technical contribution is a novel LLM agent framework that accounts for individual activity patterns and motivations, including a self-consistency approach to align LLMs with real-world activity data and a retrieval-augmented strategy

for interpretable activity generation. We evaluate our LLM agent framework and compare it with state-of-the-art personal mobility generation approaches, demonstrating the effectiveness of our approach and its potential applications in urban mobility. Overall, this study marks the pioneering work of designing an LLM agent framework for activity generation based on real-world human activity data, offering a promising tool for urban mobility analysis.

5. Ruan, J., Chen, Y., Zhang, B., Xu, Z., Bao, T., Du, G., Shi, S., Mao, H., Li, Z., Zeng, X., & Zhao, R. (2023, August 7). *TPTU: large language model-based AI agents for task planning and tool usage*. arXiv.org. <https://arxiv.org/abs/2308.03427>

Abstract: With recent advancements in natural language processing, Large Language Models (LLMs) have emerged as powerful tools for various real-world applications. Despite their prowess, the intrinsic generative abilities of LLMs may prove insufficient for handling complex tasks which necessitate a combination of task planning and the usage of external tools. In this paper, we first propose a structured framework tailored for LLM-based AI Agents and discuss the crucial capabilities necessary for tackling intricate problems. Within this framework, we design two distinct types of agents (i.e., one-step agent and sequential agent) to execute the inference process. Subsequently, we instantiate the framework using various LLMs and evaluate their Task Planning and Tool Usage (TPTU) abilities on typical tasks. By highlighting key findings and challenges, our goal is to provide a helpful resource for researchers and practitioners to leverage the power of LLMs in their AI applications. Our study emphasizes the substantial potential of these models, while also identifying areas that need more investigation and improvement.